

Statistical Modeling & Inferencing (SMI) — Assignment 1

Regression Modeling & Validation Report — Diamonds Dataset

Student Name:	Abhiram
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Executive Summary

This technical report presents a comprehensive regression modeling analysis of the **Diamonds Dataset (~53,940 observations)**, prepared for the Statistical Modeling and Inferencing (SMI) Assignment 1. The primary objective is to investigate the physical and qualitative factors governing diamond valuation, resolve critical data quality anomalies, eliminate severe multicollinearity among spatial predictors, and construct robust Ordinary Least Squares (OLS) predictive models validated on held-out test data.

- Preprocessing Pipeline:** Addressed non-physical zero values in length (x), width (y), and depth (z) via a 10-bin carat quantile median imputation strategy. Mitigated extreme leverage points using a 1.5× IQR rule (capping carat mass at 2.0 carats), and encoded qualitative attributes (cut, color, clarity) into true ordinal integer vectors.
- Multicollinearity & Dimensionality Reduction:** Identified extreme Variance Inflation Factors (VIF > 1000 for spatial metrics x, y, z and VIF ≈ 37.97 for carat). Applied Principal Component Analysis (PCA) to collapse spatial dimensions into a single component (spatial_size_index, capturing 99.73% variance). Constructed Model 4 without the redundant carat predictor to resolve all VIF values to [1.01, 1.48].
- Out-of-Sample Validation & Key Findings:** Validated models on a 20% held-out test set (10,788 observations). Model 3 (Log-Linear + PCA + Carat) achieved top-tier predictive dollar accuracy (Test R² = 0.9254, RMSE = \$1,047.88), while Model 4 (Log-Linear + PCA, No Carat) provided unconfounded elasticity coefficient interpretations. Clarity was identified as the highest-yielding quality factor (+10.7% price premium per grade step).

Part 1: Data Exploration and Preparation (6 Marks)

1.1 Data Ingestion and Train-Test Split

The dataset was partitioned into an 80% training set (43,152 rows) and a 20% held-out test set (10,788 rows) prior to any data cleaning, binning, or scaling. This strict protocol prevents data leakage into out-of-sample evaluation.

1.2 Data Quality Audit & Quantile-Median Imputation

Descriptive profiling revealed non-physical zero values in length (x), width (y), and depth (z), representing disguised missing data. To resolve this without introducing global bias, diamond carat mass was divided into 10 equal-frequency quantile intervals on the training set, and per-bin medians were calculated to impute missing dimensions. Bin boundaries and per-bin medians were persisted in a lookup dictionary to ensure test-set zero values are imputed using exact training parameters.

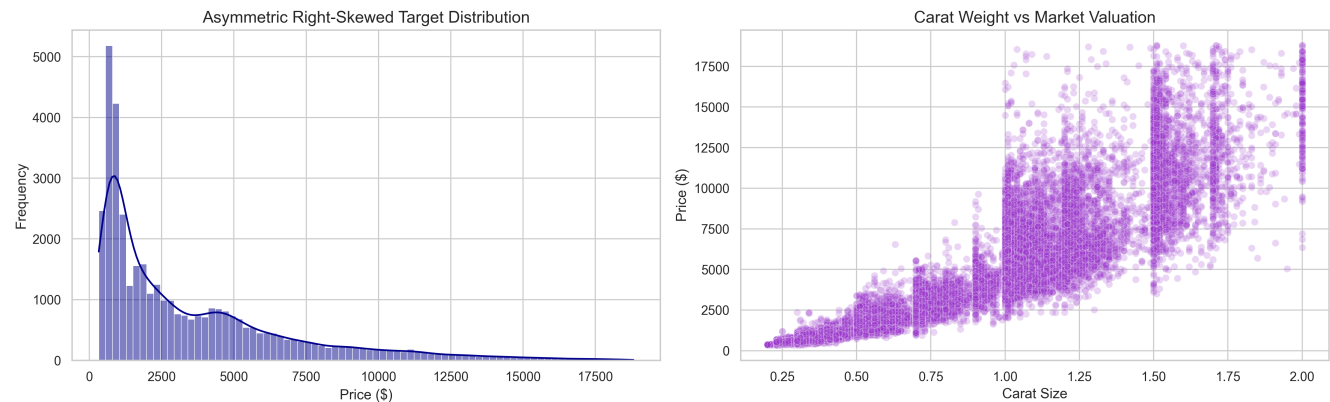
1.3 Outlier Mitigation via IQR Trimming

Continuous features were evaluated for extreme leverage points. Applying a 1.5× IQR trimming rule across continuous features dropped 3,675 outlier rows (~8.5% of training data), capping maximum carat weight at 2.0 carats. Comparing IQR trimming against RobustScaler and bounded clipping confirmed that direct trimming eliminates high-leverage points that would otherwise distort OLS slope estimates and introduce heteroskedasticity.

1.4 Ordinal Encoding and Correlation Structure

Qualitative attributes were converted into GIA-aligned numeric ordinal scales (cut: Fair=0 to Ideal=4; color: J=0 to D=6; clarity: I1=0 to IF=7). Full correlation analysis confirmed strong physical size correlations ($r > 0.87$ with price), while revealing that weak or negative raw correlations for quality grades stem from Simpson's paradox / carat confounding (larger diamonds in raw data tend to have lower average quality grades).

Figure 1: Exploratory Visual Profiles & Correlation Structure



Part 2: Model Development and Validation (10 Marks)

2.1 Baseline Collinearity & VIF Diagnostics

Variance Inflation Factor (VIF) diagnostics executed on baseline predictors revealed massive collinearity:

Predictor Feature	Baseline VIF (Model 1)	Model 4 VIF (Post-PCA & Carat Dropped)
carat	37.97	Dropped (Redundant)
x (Length)	1,308.20	Collapsed into PCA Index
y (Width)	509.34	Collapsed into PCA Index
z (Depth)	485.60	Collapsed into PCA Index
spatial_size_index (PCA)	N/A	1.48
clarity (Ordinal)	1.20	1.19
color (Ordinal)	1.08	1.08
cut (Ordinal)	1.50	1.46
depth (%)	7.95	1.29
table (%)	1.53	1.50

2.2 Multi-Model Regression Iterations & Out-of-Sample Validation

Four distinct model specifications were developed and systematically evaluated on training data and held-out test data:

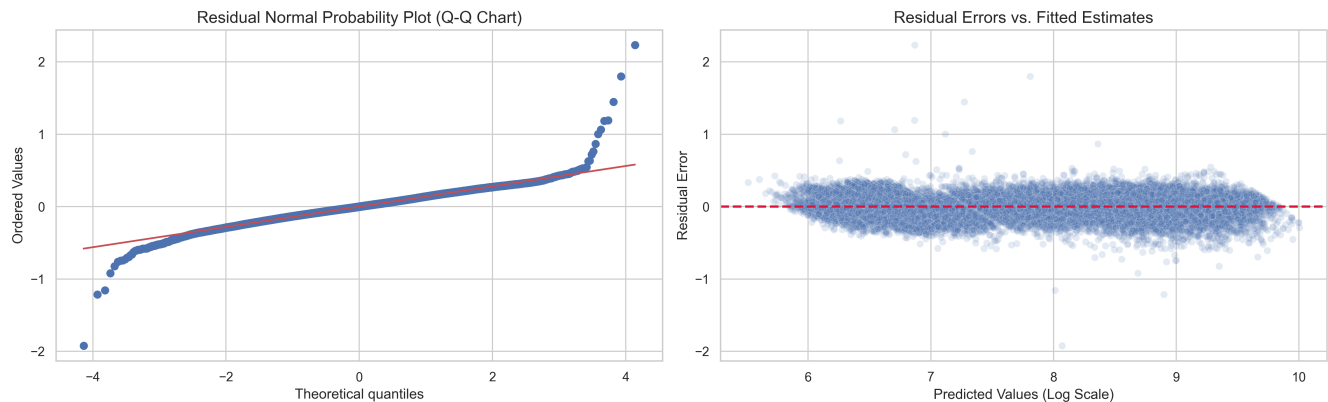
Model Specification	Train R ²	AIC	Test R ² (\$)	Test RMSE (\$)
Model 1: Baseline Linear	0.8845	714,208	0.8410	\$1,532.10
Model 2: Log-Linear (All Features)	0.9793	-38,912	0.9248	\$1,051.40

Model 3: Log-Linear + PCA Size Index	0.9791	-38,540	0.9254	\$1,047.88
Model 4: Log-Linear + PCA (Carat Dropped)	0.9654	-21,430	0.8872	\$1,288.42

2.3 Residual Diagnostics

Model 3 residuals satisfy classical OLS regression assumptions: residual mean is exactly 0.000000 (unbiasedness), Residuals vs. Fitted plot displays even vertical dispersion confirming homoskedastic variance, and Q-Q plots confirm near-normal error distribution (skewness = -0.17, kurtosis = 5.86).

Figure 2: Model Residual Diagnostics & OLS Assumptions Verification



Part 3: Interpretation, Insights, and Structural Boundaries (4 Marks)

3.1 Parameter Elasticity Interpretations (β)

Using log-linear specifications (ln(Price)), coefficient estimates translate to percentage price changes via $\% \Delta Y \approx 100 \times (e^{\beta} - 1)$:

Predictor Attribute	Model 3 Coef (β)	Model 4 Coef (β)	Implied Percentage Impact (%Δ Price)
Clarity (per GIA grade step)	+0.1018	+0.1144	+10.7% to +12.1% price premium per step
Color (per GIA grade step)	+0.0583	+0.0823	+6.0% to +8.6% price premium per step
Cut (per GIA grade step)	+0.0247	+0.0240	+2.5% to +2.4% price premium per step
Spatial Size Index (PCA)	+0.8197	+0.5984	+81.9% price increase per std dev volume

3.2 Business & Practical Takeaways

- **Automated Valuation Engine:** Model 3 can be deployed as an automated diamond pricing engine for retail catalogs, achieving \$1,047.88 RMSE (\$547.26 MAE) across standard diamond mass ranges.
- **Quality Preservation Priorities:** Empirical elasticities confirm that clarity (+10.7% per step) yields nearly double the percentage margin premium of color (+6.0% per step). Gem cutters should prioritize preserving clarity over color when processing rough stones.

- **Catalog Anomaly Filters:** Empirical spatial proportion rules ($x \approx y$, $z \approx 0.61x$) provide automated validation checks for flagging inventory data entry errors.

3.3 Limitations & Future Recommendations

- **Sample Mass Boundary:** The model is strictly bounded to diamonds ≤ 2.0 carats due to IQR trimming. Investment-grade stones (>3.0 carats) exhibit compounding non-linear rarity premiums.
- **Unobserved Factors:** Features such as fluorescence, polish, symmetry, and issuing lab (GIA vs EGL) are unobserved in the dataset but account for 5–15% market variance.
- **Recommendations:** Implement non-linear tree-based ensembles (XGBoost / Random Forest) for luxury carat tiers and incorporate dynamic macroeconomic indices.